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Applying O&G ALS Technology to Enable Worldwide Food Security and a Greener Future by Improving Boron Production Efficiency in a Mining Application Project Developed in California

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Abstract

A common mindset considers mining sector as one of the dirtiest economic activitiescurrently killing the planet. In similar way, the Oil and Gas (O&G) industry has been by far persecuted ascontaminant and sustainably irresponsible. Contradictory to such bias, this paper aims to summarize theresults of utilizing Artificial Lift Systems (ALS) technology from O&G in mining applications to improve theefficiency of boron production in a project developed in Newberry Springs, California. USA.

This multiyear project was focused on understanding the miningapplication from ALS perspective, screening lift options, and finally select the most suitable solutions toimprove boron production efficiency. Boron is produced from acid injection in a well and then soaked forcertain hours; Significant amount of non-pressurized boric-acid, borates and water are generated at pay-zone; with an important concentration of chlorides, generating a highly corrosive bottomhole conditionwith no static-pressure to lift these fluids to surface. ALS becomes a reliable solution for flow assurance inmining. Based on well condition, Progressing Cavity Pump (PCP) system was selected for this project.

A wide variety of test were conducted to select the proper materialsfor the PCP system, including elastomer compatibility test, tube corrosion test, elastomer/tube bondingtest, immersion elastomer/tube test, sucker rod / tubing wear test, torque anchor gripping test, etc. Inaddition, alternative materials and coatings were explored on components like PCP rotor, sucker rod stringand surface equipment. The tests were conducted at downhole temperature criteria to simulate operational conditions.

O&G industry know-how helped to enable food security through the deployment of a custom-madeinsert-PCP system that allows effective boron extraction in a mining application in California. For thisspecific case, the insert-PCP increased production efficiency due to the combination of an abrasion-resistant rotor and a soft elastomer in the stator. PCP system equipped with a conventional sucker rodstring made of hastelloy C276 and paired with an energy-efficient permanent magnet motor drive headand is part of a proof of concept to apply traditional O&G artificial lift technology to effectively mineboron, a material deemed critical for the global energy transition.

Introduction

For more than 100 years, the mining activity has been taking place in USA, offering different industrial minerals, rare-earth elements, clays, limestone, gypsum, and salt. The metals include gold, silver, iron, and coper (1. California's Mineral – Department of Conservation, 2020). Finally, USA extracts construction aggregates including sand, gravel, and crushed stone. These minerals are available in the Mojave Desert located in the west side of USA, more specifically in the state of California, in the Newberry Springs area.

Newberry Springs is an unincorporated community in the western Mojave Desert of Southern California, located at the foot of the Newberry Mountains in San Bernardino County, California, USA. The community is located at the east side of Barstow, west side of the Mojave National Preserve, and south of Death Valley National Park. Figure 1. Shows California State, USA, Physical map with Newberry Springs location. The town is approximately 2,000 ft (610 m) above sea level. The area is irrigated by the Mojave Aquifer, the largest aquifer in the Western United States. Paradoxically, Newberry Springs is a typical desert oasis (2. Newberry Springs, California, 2009) with ancient volcanic rock formations, lava beds, sand dunes, and several minerals, including Boron (3. Mining.com, 2021).

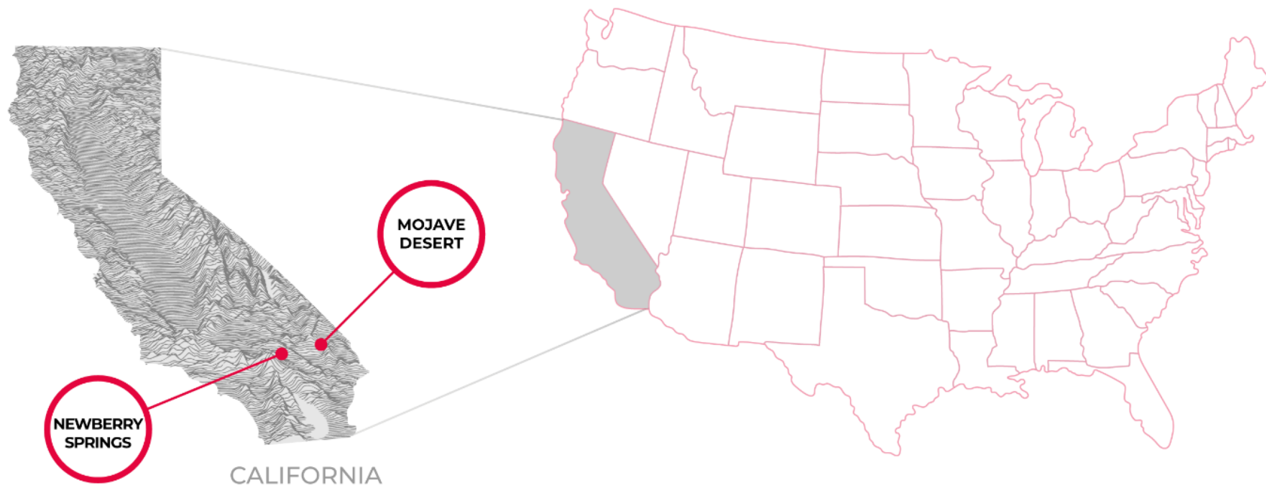


Figure 1—California State, USA, Physical Map with Newberry Springs location.

What is Boron?

Boron is the 5th element on the Periodic Table with a powerful combination of physical properties, including hardness, light weight, and heat resistance. The Boron is naturally found as solid. The name derives from the Arabic buraq for "white". Although its compounds were known for thousands of years (4. CIAAW, IUPAC, 2020), boron was discovered by Joseph-Louis Gay-Lussac and Louis-JaquesThénard, French chemists, and independently by Sir Humphry Davy, an English chemist, in 1808. They all isolated boron by combining boric acid (H_3BO_3) with potassium. Today, boron is obtained by heating borax ($Na_2B_4O_7 \cdot 10H_2O$) with carbon, although other methods are used if high-purity boron is required (5. Thomas Jefferson National Accelerator Facility - Office of Science Education, 2021). Boron is one of the most versatile materials in the world. Therefore, it can be found in over 300 applications that improve the current society's lifestyle. Some of the applications where Boron is utilized in our daily life, include fertilizers, Permanent Magnet Motors (PMMs), aerospace ceramics, wind turbines, cosmetic products, nuclear reactors, detergents, Electric Vehicles (Evs), body/vehicle armor, solar panels, gypsum, wood treatments, glass, fire retardants, and pharmaceuticals.

The USA government has designated Boron as a strategic commodity due to its unique properties and criticality to national security and clean energy goals, especially as demand for these supply chains grows.

The major ores of Boron are a small number of Borate (boron oxide) minerals, including ulexite ($NaCaB_5O_9.8H_2O$), Borax ($Na_2B_4O_5(OH)_4.8H_2O$), Colemanite ($Ca_2B_6O_{11}.5H_2O$) and Kernite ($Na_2B_4O_6(OH)_2.3H_2O$). These minerals form when Boron-bearing waters percolate into inland desert lakes and evaporate, leaving layers of borates, chlorides, and sulfates. These minerals are referred to as evaporite minerals. Very large deposits of evaporite Boron minerals are found in the USA (especially California), Turkey, Chile, and Argentina. Less-important deposits occur in Iran (formerly called Persia), and elsewhere.

Other geographical areas producing Boron, as follow: North America (Canada and Mexico); Europe (Germany, UK, France, Italy, and Russia); Asia-Pacific (Japan, Korea, India, Australia, Indonesia, Thailand, Philippines, Malaysia, and Vietnam); South America (Brazil, and Colombia, etc.); Middle East and Africa (Saudi Arabia, UAE, Egypt, Nigeria, and South Africa). Turkey is the largest producer of Boron, with a market share about 45%. It was followed by North America with 30%. The top 3 mining companies with the 75% of the market share are in California (6. Food Industry Insights, 2023). Figure 2 shows the several projects and companies extracting Boron worldwide:

Figure 2—Global Boron Extraction Map.

In summary, Boron is an element that is contributing to the food security, the development of clean energy as well as in electric transportation.

Boron Cycle

The Boron cycle can be divided into 2 specific scenarios: One scenario that happened millions of years ago, which is a result of the volcanic emissions which created what is currently well known as Mining of Boron ores and coal; as well as boron reservoirs settled not deeper than 1200 m (3937 ft). The second scenario in the Boron cycle resulted from the extraction of the boron from the mining activities in Boron ores. As a result, the produced Boron (Borates) is processed into industrial products (Fertilizers, cosmetics, glass, etc.,). Then, the fertilizers that contain boron are utilized for farming; the fruits and crops will have a content of Boron and the trees will have some boron content that when fire season comes, the Boron is expelled to the atmosphere, as well as to the water (rivers and then ocean), in parallel the boron is deposited to the ocean and rivers as rainfall to the land and to water bodies until the Boron is settled down as a sediment until is again extracted from the mining of Boron activities. And ultimately, this second scenario of the cycle is restarted again; over and over. (7. Jefferson Lab, U.S Department of Energy, 2022)

Details of Boron cycle are showed below in Figure 3. Boron in the atmosphere is also derived from soil dusts, volcanic emissions, evaporation of boric acid from seawater, biomass emissions, and sea spray. These ones are the light red arrows in this diagram. The Boron cycle is the biogeochemical cycle of Boron through the atmosphere, lithosphere, biosphere, and hydrosphere. The Boron cycle has been significantly impacted by human activity. Major impacts are coming from coal mining and combustion, oil production, emissions from industrial factories, biofuels, landfills, and mining and processing of Boron ores.

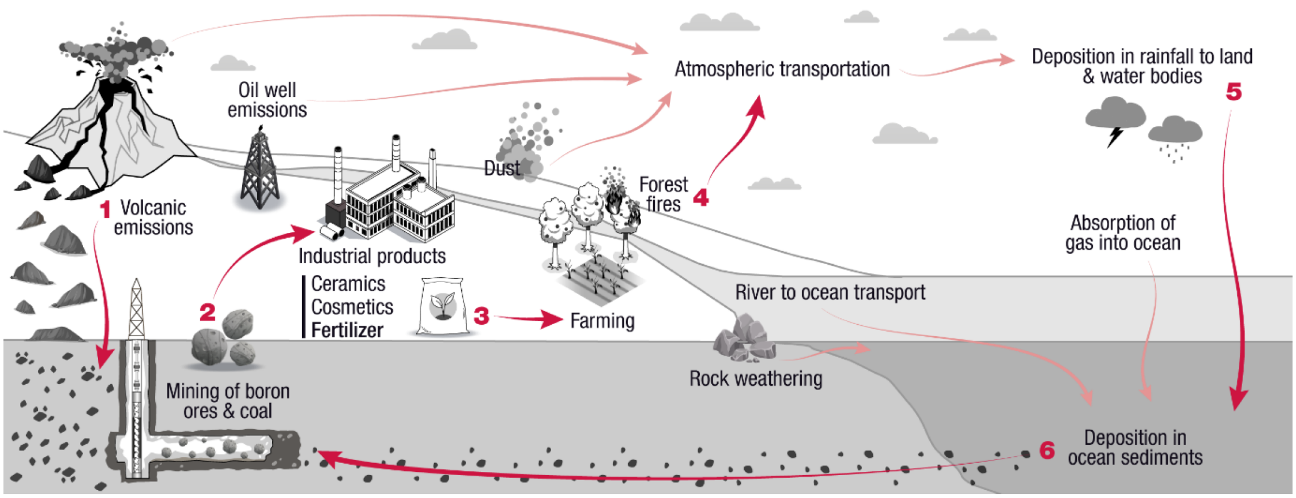


Figure 3—Boron Cycle

Boron Mining and Extraction

Most commercial borate deposits in the world are mined by open pit methods. The world's major borate operations, the Boron mine of US Borax at Boron (Kramer), CA, and the Kirka mine of Eti-bank in Turkey are huge open pit mines utilizing large trucks and shovels and front-end loader methods for ore mining and overburden removal. Orates are mined by underground methods in the Liaoning area of northeast China, at the Billie and Gerstley mines in Death Valley, CA, and in Turkey at Emet and the Simav mine at Bigadic. Borate brines are recovered at Searles Lake, CA, and in the Qinghai Basin of China; brines may also be utilized in Kazakhstan. Borate containing brines are being considered for production in South America. However, no Borates have been produced since 1991. (8. Kistler&Helvaci, 1994) Boron is produced as a result from acid injection in a well and then soaked for certain hours; Then, a significant amount of non-pressurized boric-acid, borates and water are generated at the pay zone.

Application's Details

The mining application in study is a well completion with a 177.8 mm (7 in) cemented CSG × 88.9 mm (3 1/2-in) FRP TBG, the bottom hole temperature around 50°C (122°F) with an important concentration of chlorides, generating a highly corrosive bottom-hole condition with no static pressure to lift these fluids to surface. As mentioned previously, the application is subject to hydrochloric acid (HCl) injection from surface, a soaking time to produce water and Borate through the bottom zone. The pay zone is a 60 m (200 ft) thick colemanite layer. The expected flow rate is around 63 m^3/d (≈395 bbls/d), but it can go up to 163 m^3/d (1023 bbls/d). The below diagram in Figure 4 shows the primary layout of how the chemical injection process is executed. The original Borate extraction method was by air lifting which indeed had a poor lift efficiency.

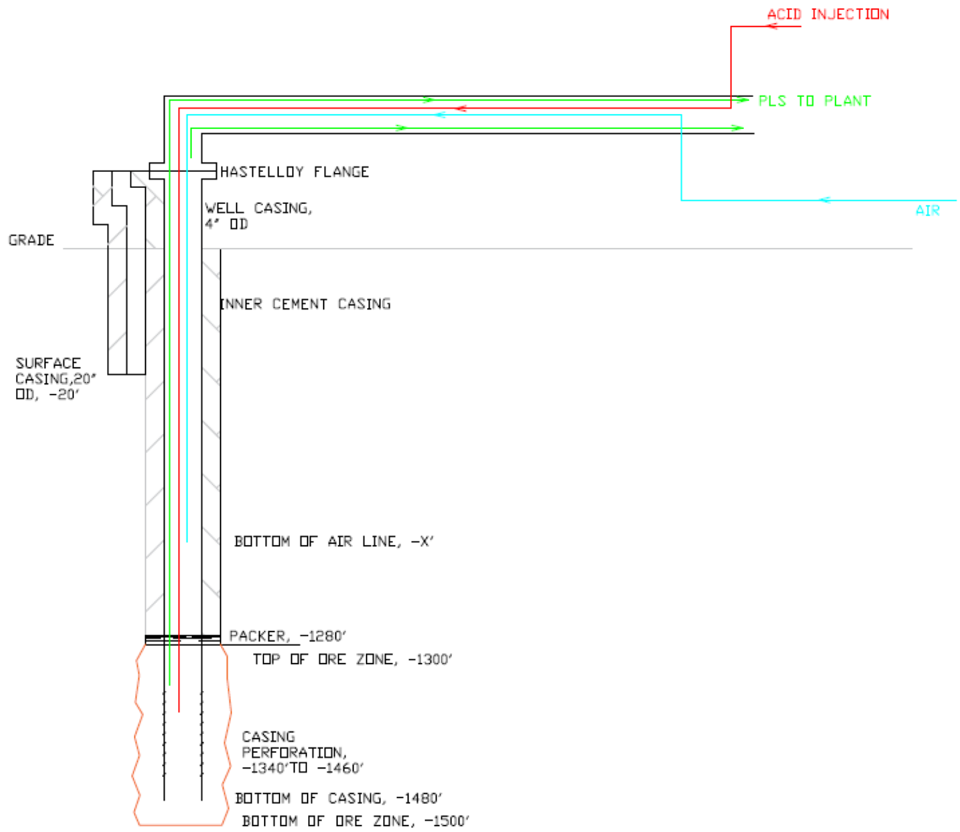


Figure 4—Original extraction method included air injection with poor lift efficiency. (5E-Advanced Materials, Inc., 2021)

Design Requirements

These are the design requirements considered for this application:

- Capable to operate reliably in highly corrosive environment.
- Retrievable solution with a rigless unit through 88.9 mm (3 1/2-in.) FRP TBG.
- Cover production flow rate expectation. [up to 163 m^3/d (1023 bbls/d)]
- Capable to consistently lift fluids from 400 m (1300 ft) to surface.
- Improve lift efficiency in the well.
- Ability to handle solids.

Technical Assessment

Considering the poor lift efficiency in the well from air lift as the extraction method, ALS becomes the ideal solution for Boron extraction in mining applications. Due to the high corrosive environment in these wells, the ALS selection will require the use of exotic material in its components.

Artificial Lift System – ALS

ALS is a method used to increase pressure within the fluid to encourage it to the surface. When the natural drive energy from the reservoir is not strong enough to push the fluid to the surface, ALS is employed to recover more production. While some wells contain enough pressure for fluid to get to the surface without any help, most do not, requiring an ALS. In fact, 96% of the O&G wells in USA require artificial lift from the very beginning. Even those wells with natural flow to the surface from the beginning, pressure depletes over time, and ALS is then required. Therefore, ALS is generally performed on all wells at some time during their production life. (9. Rig Zone, 1999)

There are several types of ALS:

- ***Reciprocating Rod Pump (RRP):*** Probably the most common type of ALS applied. Works like a piston inside a cylinder operating in a back-and-forth movement to lift the fluid from the producing zone up to surface.
- ***Progressing Cavity Pump (PCP):*** Utilizes a positive-displacement downhole pump, the pump pressurizes the fluid, allowing the fluid to navigate from the producing zone up to surface (10. Lea, 2006)
- ***Electro Submersible Pump (ESP):*** With a downhole centrifugal pump, pressurizes the fluid to navigate from the producing zone up to surface.
- ***Gas Lift (GL):*** Decreases density of the fluid in the tubing and expands gas to help the fluid lifting to surface.

A common practice is to classify the ALS into 2 categories: rod-less and rod-driven. The Figure 5 shows some of the most common ALS available in the market.

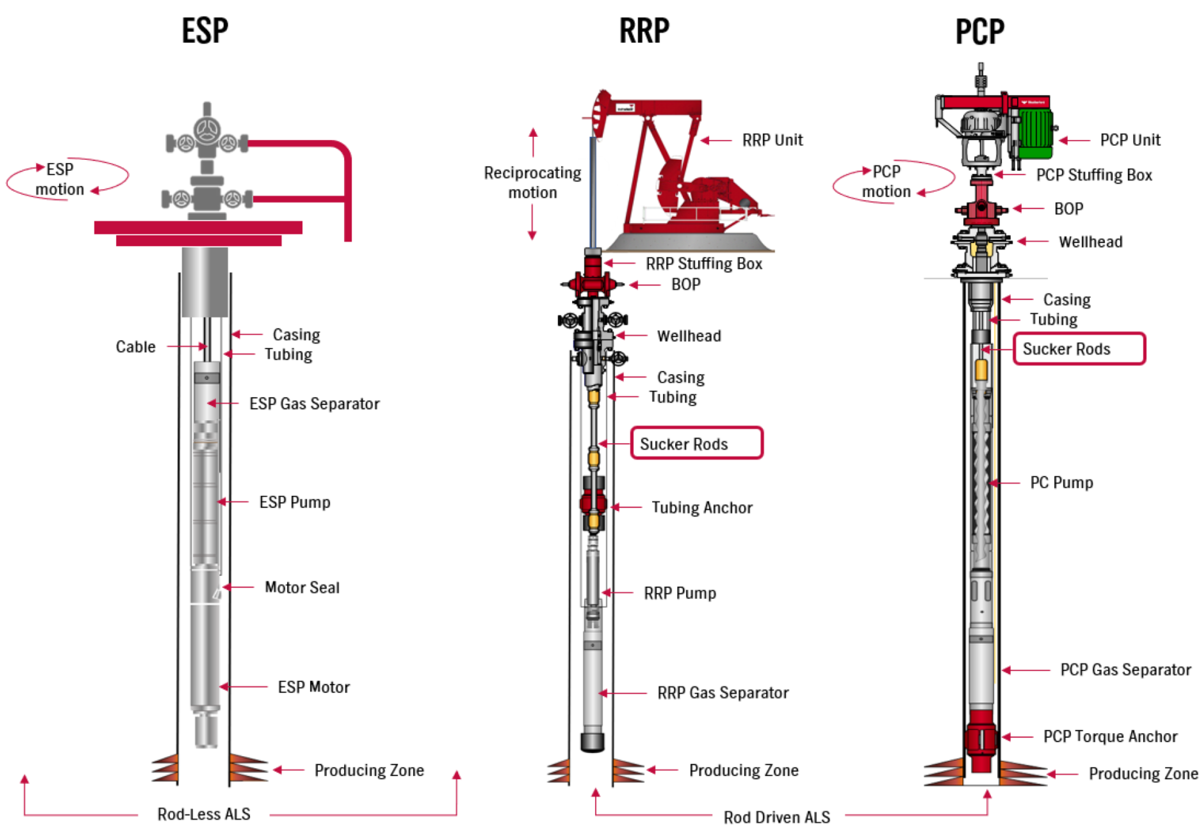


Figure 5—ALS Comparison. Rod-Less ALS vs Rod Driven ALS: ESP vs RRP & PCP. (Weatherford. 2023)

Selecting the proper ALS in an application. During the planning stage in a project focused on the development of a new application, the selection of the ALS to be installed in the field would be one of the most critical decisions. Aspects such as flow rate, produced fluid viscosity, equipment installation simplicity and its capability for redeployment must be evaluated to pick the most cost-effective option for this application.

ALS Selection Matrix. A proper selection of the ALS to be deployed in each application should be done consciously but sometime could be complex. Weatherford has developed a well-defined guide through an ALS Se-lection Matrix to select the most suitable form of lift based on the well condition. The main objective of this matrix is to holistically help with the ALS selection process, to increase the success rate when deploying the selected ALS. Table 1 contains some parameters to be considered when evaluating ALS options.

Table 1—ALS Selection Matrix. (Weatherford. 2019)

	PCP	RRP	ESP	JET PUMP	GAS LIFT	PLUNGER LIFT	FOAM LIFT
	EXTERNAL ENERGY ASSISTED			LEVERAGES NATURAL ENERGY			
Comment	Low cost, high flow, insert pump	Med cost, lower end of flow rate, material concerns, tubing pump.	High cost, high flow, material concerns	Surface equipment material req's. Dilute acid too quickly	No natural BHP. Pressure exceeding FBG limits	Production rate too low	Production rate too low
Max Depth [ft]	8,600	16,000	15,000	20,000	18,000	19,000	22,000
Max Volume [bpd]	5,000	5,000	60,000	35,000	75,000	200	500
Max Temp [°F]	300	550	550	550	450	550	400
Corrosion Handling	Fair	Good to Excellent	Good	Excellent	Excellent	Good to Excellent	Excellent
Servicing	Service rig or Workover rig		Workover rig	Hydraulic retrieve or slickline	Wireline or Workover rig	Slickline	Capillary unit

The Run Life (RL) and Mid Time Between Failure (MTBF) as well as financial indicators like Lifting Cost and Return on Investment (ROI) will be positively affected by the proper selection of the ALS. In the specific case of Newberry Springs field, PCP can operate within the range of maximum depth, maximum volume, maximum temperature, and fluid gravity requested for this application.

After a revision of the reservoir / well condition, the PCP system was selected because of its simplicity, ability to handle solid particles, the lack valves and reciprocating parts and the use of elastomer combined with helical shaped rotor.

RRP was the initial ALS considered for this application; however, since this ALS has many components, a significant Research & Development (R+D) investment would have to be made, making it a non-cost-effective application due to the high number of changes in the metallurgy.

ESP was evaluated but due to the history of damaged impellers when dealing with high percentage of solid particles in this application; this ALS form was not selected as a solution.

Based on the well condition, a PCP system was selected as the most suitable for this project. More details about PCP are listed below.

Progressing Cavity Pump - PCP

The PCP system is one of the most recent forms of ALS. It was initially developed in the late 1920s and patented in the 1930s by Rene Moineau (11. Moineau& Joseph, 1932) and used for highly vis-cous applications since 1940. However, the initial downhole application was in the Canadian heavy oil fields where wells produced high viscosity sand-laden fluids that had proven difficult to economically produce with the time's existing oilfield artificial lift systems. By 1990, PCP had its vast expansion through Cold Heavy Oil Production with Sand (CHOPS) technology in Alberta, Canada. (12. Dusseault, 2002.)

The PCP system has several components that can be divided into two main equipment types: The downhole and surface equipment. The downhole equipment has, along with other pieces, the PC pump, which consists of two main components: A double-helical, elastomer-lined tube installed be-low the tubing string; known as the Stator, this is the stationary component; the elastomer is select-ed depending on several operational parameters such as bottom-hole temperature, sand content tolerance, aromatics CO_2 and H_2S content. The second component is a single helical steel bar with an abrasion-resistant coating, typically hard chrome, known as the Rotor, which is the rotational component connected to a rod string. The rotor turns inside the stator, and this combination corresponds to the PC pump and creates sealed cavities where fluid is gaining pressure progressively while traveling through one cavity to the other from the pump intake to the pump discharge section. The result is a non-pulsating flow with a discharge rate proportional to the rotor's rotational speed and the pump displacement (13. Ariza et al., 2020). See Figure 6.

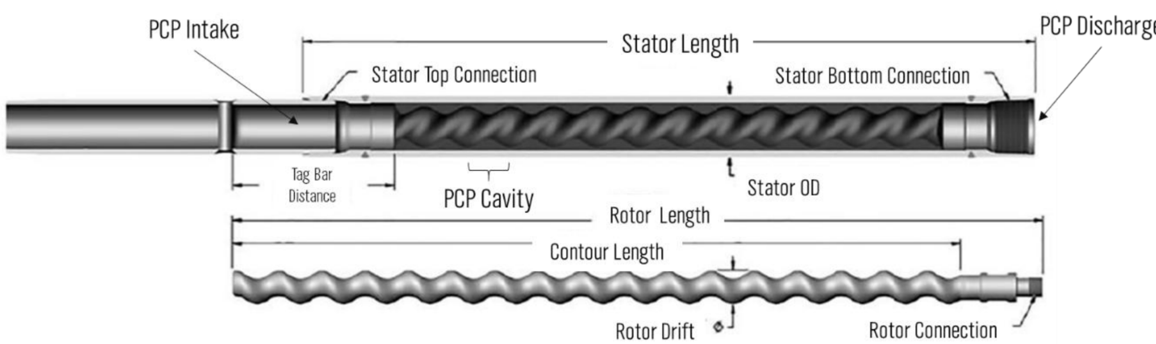


Figure 6—PC Pump components

The surface equipment transfers the rotational speed and torque to operate the PCP system through the electric motor. Then the Belts and Sheaves rotate and transmit the power given by the electric motor through rotational energy. The Polished Rod Clamp then transmits the rotational energy provided by the belts/sheaves to the Polished Rod. The surface drive should also be strong enough to suspend the rod string and support the axial load. Lastly, the stuffing box is required to act as a perfect seal to avoid any leakage on the surface.

Over the years, the PCP system has become the most suitable option to control high sand rates and sand production problems in reservoirs with no "matrix" due to non-consolidated formation. Lastly, the PCP system has outstanding performance in high slurry formation flow resulted from a mixture of sand-oil-gas-water. (14. Martinez et al., 2021). PCP systems have lower CAPEX and lower operating cost due to the fewer and smaller components in this system. PCP is a highly efficient system when handling high sand cuts, and heavy-highly viscous fluids. This is the result of the rotor-stator combination. It also has excellent resistance to abrasion, because of the combination of an abrasion-resistant coating on the rotor and a soft elastomer surface in the stator.

Insertable-PCP (I-PCP)

I-PCP is a category of PCP in which the entire pump assembly is installed via the rod string and landed inside the tubing, enabling the pump to be pulled and rerun by the rod string. I-PCPs offer many advantages downhole, including, primarily reducing downtime related to costly and time-consuming tubing pulls, which occur when changing worn-out conventional pumps, replacing dam-aged pumps, or switching to a different pump size/configuration when downhole pumping requirements change. I-PCPs also eliminate the need to pull and rerun downhole instrumentation. The I-PCPs can be installed into different TBG sizes 73 mm (2 7/8-in.), 88.9 mm (3 1/2-in.), 114.3 mm (4 1/2-in.), and 139.7 mm (5 1/2-in.). As the I-PCP travels through the TBG string to the desired landing depth, no Work Over Unit is required for deploying or retrieving the I-PCP to the production zone (15. Weatherford. 2016); this condition becomes in an advantage of I-PCPs over conventional PCPs; an I-PCP can be deployed in few hours without pulling any TBG string; in contrast, the installation of a conventional PCP will require to pull out TBG string with a big Work Over Unit and the job could take minimum 1 day due to the pulling and running of TBG string together with the conventional PCP.

A rigless unit, such as a service unit, CORIG and/or Flush-by unit, wireline unit, or a coiled tubing unit can be used to pull and run the I-PCP. Note that because the entire pump assembly must fit inside the production tubing, there is a size limitation, which reduces the maximum flow rate. Therefore, I-PCP systems could have lower flowrates than conventional tubing deployed systems.

The I-PCP can be installed using two separate methods: The conventional method is by a Pump Seating Nipple (PSN) in the tubing string and a corresponding set of seating rings in the pump assembly. The I-PCP system is run and deployed by lowering it to the target position (PSN or anchor landing depth). Retrieval of the I-PCP is done by pulling the rod string upwards to remove the seating rings from the PSN or unseating the anchor if it was used. Once at surface the I-PCP can be inspected and normally redressed, if required, on the well site and rerun. I-PCP systems have ap-plication for a wide range of requirements depending on displacement and lift capacity (16. Jimenez et al., 2015).

I-PCP Components. The I-PCP can be installed with a top hold down or a bottom hold down configuration. The components of the top hold down I-PCP configuration are showed below in the Figure 7. The I-PCP has a top seating mandrel (Num. 4) which main function is to house the floating ring (Num. 6) used for installing and retrieving the insertable PC pump at the desired landed depth. The top seating mandrel also contains the seating rings (Num. 3) which creates a seal when in contact with the inner side of the Pump Seating Nipple (PSN). The floating ring located inside the top seating mandrel prevents the rotor from completely being disengaged from the I-PCP assembly. The Arrowhead tip located at the bottom of the rotor (Num. 14) has been designed to meet the inner shaped of the floating ring and prevents the rotor from being separated from the I-PCP assembly while installing and retrieving the I-PCP. An insertable PCP anchor is installed below the tag bar (Num. 17) and is pre-vents the rotation and axial movement of the stator when a PSN is not present in the well.

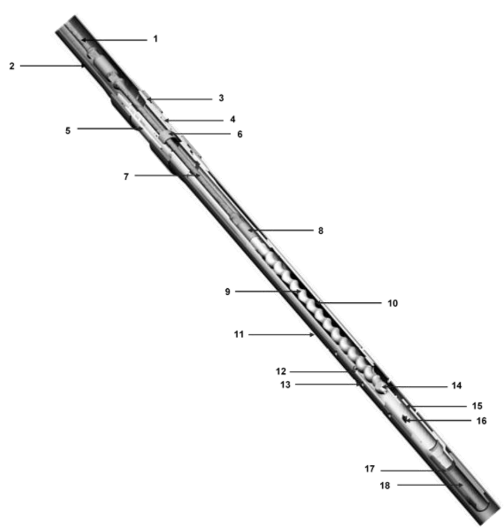


Figure 7—I-PC Pump components (Weatherford. 2019)

Test for Material Selection

Considering the HCl to be injected in these wells to produce the Borates; several metallurgies were tested resulting on the selection of some exotic materials to manufacture the different components for the project.

It was also required to modify the standard manufacturing procedures to offer a reliable ALS to prove the thesis: "Deploy O&G ALS technology in mining applications to improve the Borate extraction efficiency."

Aligned to the more efficient principle that drives this project, the installation of a highly efficient PMM will help to keep a suitable amount of torque even when this ALS system will be operating at a lower speed. Some of the elements in the PMM design are earth-rare minerals which are indeed, extracted from mines in California.

This contributes to the circular economy trend we are getting by these days and is so interesting how the O&G sector is utilizing the ALS technology to get a higher boron extraction rate compared to what the mining industry has as of today.

The PCP style selection (conventional PCP or I-PCP) relied on the client's limitation of getting a rig on site to pull out TBG string in case of any damage in the well. Considering the flexibility of the I-PCP over the conventional PCP when retrieving the TBG string, (I-PCP can be pulled out with smaller rigless unit, giving more options of pulling units available when required), an I-PCP was selected for this project.

The operator requested FRP TBG as the preferred material for this application because of the highly corrosive environment, the FRP TBG specifications are showed in Figure 8. TBG inner diameter (ID), TBG drift diameter, and maximum FRP TBG make up torque capacity were probably the most critical parameters to be considered for the I-PCP selection. The maximum FRP TBG make up reduces the torque available in the system, as well as the breakaway and operational torque in the pump.

Series 1500

1500 psi (10.3 MPa)

Thread Size	Product Code	Nominal Dimensions								Connection Diameter				Tensile Rating[3]		Collapse Rating[3]	
		Inside Diameter		Drift Diameter		Outside Diameter		Weight[1]		IJ[2]		T&C[2]					
in		in	mm	in	mm	in	mm	lbs/ft	kg/m	in	mm	in	mm	lbs	kgs	psi	MPa
3 ½	C1529	2.94	74.7	2.82	71.6	3.39	86.0	2.12	3.2	4.51	114.6	4.5	116.7	18,000	8,165	1700	11.7
3 ½-DH	C1529LA	2.94	74.7	2.82	71.6	3.39	86.0	2.17	3.2	4.66	118.4	-	-	18,000	8,165	1700	11.7

Figure 8—FRP TBG specs 88.9 mm (3 ½-DH). (NOV Fiber Glass Systems, 2021)

Briefly, the breakaway torque corresponds to all the required torque to fill out all the PC pump cavities and overcome the friction between the PCP Rotor and PCP Stator. This condition is common when starting up a pump.

The breakaway torque will take place in several scenarios like:

- When passing from a static condition to a dynamic condition
- When starting a well because of any energy interruption
- When solid particles are passing through the I-PCP (a more heavier weight condition)

Given the previous scenarios, the breakaway torque value would be as high as 2.5 times the operational torque. As showed in the graph below, this torque value would be required in seconds and is highly dependent on the produced fluids and formation features (muddy, sandy, clay, etc.). The Max FRP tubing make up torque was 300 ft*lbs. Then, it was required to find other geometries that can have a balance between the required production and an operational torque not greater than 300 ft*lbs. Therefore, the geometry should have the ability to manage a breakaway torque and the operational torque lower than Max

FR make up torque. The I-PCP model selected was the 30-700 XL AY. Torque simulation along the time is showed on Figure 9.

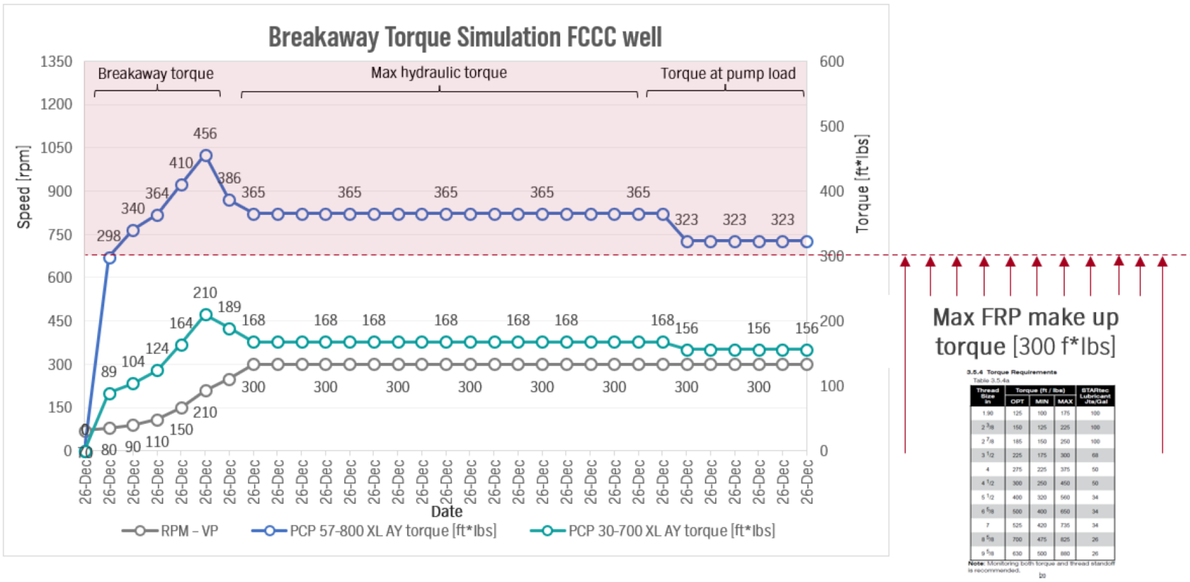


Figure 9—I-PCP Breakaway torque vs maximum FRP tubing make up torque

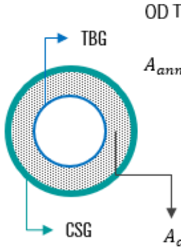
The I-PCP model 57-800 XL AL Torque simulation has been added to the graph just for educational purposes to show that its torque was greater than maximum FRP TBG make up torque (red section). Therefore, this PCP model was discarded from the PCP system final configuration.

As HCl was expected to be injected in the well to keep the reservoir soaking for around 8 hours to generate the boric acid; it was required to calculate the annulus section between the CSG ID – TBG OD to get the total volume to be drained while the I-PCP model 30-700 XL AY was in operation. In addition, it was required to calculate the time required to drain this volume in the annulus so a maximum time for I-PCP in continuous operation could be set. The Figure 10 shows the Volumetric Capacity Calculations:

$$A = \pi r^2 \rightarrow A = \pi \left(\frac{D}{2}\right)^2$$

$$A_{annulus\ CSG-TBG} = A_{ID\ CSG} - A_{OD\ TBG}$$

ID CSG: 5.5 in. → ID CSG/2 = 2.75 in.
OD TBG: 3.39 in.→ OD TBG/2 = 1.695 in.



$$A_{annulus\ CSG-TBG} = \pi \left[\left(\frac{ID_{CSG}}{2}\right)^2 - \left(\frac{OD_{TBG}}{2}\right)^2\right]$$

$$A = \pi r^2 \rightarrow A = \pi \left(\frac{D}{2}\right)^2$$

$$A_{ID\ CSG} = 23.7584\ in^2$$
$$A_{OD\ TBG} = 9.0259\ in^2$$
$$A_{annulus\ CSG-TBG} = 14.7325\ in^2$$

IF PLD = 1300 ft:

$$V_{annulus\ CSG-TBG}\,[ft] = A_{annulus\ CSG-TBG} * PLD$$

Figure 10—Volumetric Capacity Calculations

The expected time to drain the calculated annular section between CSG-TBG is 76 min; assuming a PCP model 30-700 XL AY operating at 300 RPM with a PCP volumetric efficiency around 80%, specific calculations listed below, in Table 2:

Table 2—Time (min) to drain for calculated annular section between CSG ID – TBG OD

Case	TBG 3 1/2 in. I-PCP 30-700 446FPD
Pump Description	WFRD (30-700 XL AY)
Pump Displacement (m^3/D/RPM)	0.296
Pump Speed (RPM)	**300**
Volumetric Efficiency (%)	80
Fluid Flow Rate (bbls/d)	**446.83**
Fluid Flow Rate (bbls/hr)	**18.588**
Time to drain V annulus CSG-TBG (hr)	1.27
Time to drain V annulus CSG-TBG (min)	**76.46**

Elastomer Compatibility Test (ECT) for Elastomer Selection

The purpose of performing an elastomer compatibility test is to evaluate the comparative ability of elastomers to withstand the effects of produced fluids, chemicals, etc. It is performed routinely to help in selection of PCP elastomers best suited for the applications, as well as finding out their swell level to determine stator/rotor sizing.

Elastomer tensile samples are positioned into test cells along with the testing fluid. The test cells would then be placed into either an autoclave or aging block oven and tested at a specified temperature, pressure if applicable, and duration.

The elastomer evaluation was conducted with a mixed simulation fluid containing 5% HCl and 5% H_3BO_3. In total, 4 elastomer samples were selected including: 3 nitrile elastomers (NBR) and 1 hydrogenated elastomer. The test conditions are showed in Table 3.

Table 3—Test Conditions for Elastomer Compatibility Test (ECT)

Test Parameters / Test	Test Conditions	
	Test Temperature	ECT Time
Test 1	122°F (50°C)	168 hrs
Test 2	176°F (80°C)	168 hrs

As soon as the ECT is complete, a variety of tests will be performed on the elastomer samples to identify the effects the testing fluid had such as change in mass, volume, hardness, tensile strength, elongation, and modulus. The ECT results showed good performance with the hydrogenated elastomer.

Lastly, considering the corrosive environment where the downhole PCP system will be soaked after the HCl injection, it was required to select a strong enough material that could deal with this critical condition; this material es Hastelloy C276.

Hastelloy C276 is a Nickel-chromium-molybdenum wrought alloy that is considered the most versa-tile corrosion resistant alloy available. This alloy is resistant to the formation of grain boundary precipitates in the weld heat-affected zone, thus making it suitable for most chemical process applications as welded condition. Hastelloy C276 also has excellent resistance to pitting, stress-corrosion cracking and oxidizing atmospheres up to 1900°F. Hastelloy C276 has exceptional resistance to a wide variety of chemical environments (17. Hpalloy, 2017). Several components in the PCP system were made of Hastelloy C276: Polished Rod, Sucker Rod, Couplings, PCP stator, among others.

Results

The PCP system in the 5E-Advanced Materials, Inc well was installed on December 12th, 2023. The well started its intermittent operation on January 2nd, 2024, with no high torque alarm, nor conventional sucker rod disconnection event. This is a milestone for underground mining applications in California, proving the concept that the O&G ALS technology can be effectively implemented for Boron extraction in a continuous operation of the PCP system in the 5E-Advanced Materials, Inc well.

Conclusions

- The implementation of O&G ALS technology concepts has been proven in a mining application to produce Boron, in Newberry Spring California. This becomes a milestone due to the use of this technology while enabling food security and a greener future through the Boron production in this area.
- During the planning stage in this new application, the selection of the ALS is one of the most critical decisions. Aspects such as flow rate, produced fluid viscosity, equipment installation simplicity and its capability for redeployment must be evaluated to pick the most cost-effective option for the application.
- The basic understanding of downhole environment and well conditions were key factors for the se-lection of the PCP system as the most suitable ALS for this application. The PCP was selected because of its simplicity, ability to solid particles and the lack valves and reciprocating parts.

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Nomenclature

ALS	Artificial Lift Systems
AY	Arrowhead Insert
CAPEX	Capital Expenses
CSG	Casing
ECT	Elastomer Compatibility Test
ESP	Electrical Submersible Pumping
EU	European Union
EV	Electric Vehicles
FRP	Fiberglass
GL	Gas lift
HCl	Hydrochloric acid
ID	Inner Diameter
I-PCP	insertable-PCP
LCD	Liquid Crystal Display
MTBF	Mean Time Between Failure
NBR	Nitrile Elastomer
OD	Outer Diameter
O&G	Oil and Gas
PCP	Progressing Cavity Pumping
PMM	Permanent Magnet Motor
PMM-PCP	Permanent Magnet Motor for PCP

PSN	Pump Seating Nipple
RL	Run Life
ROI	Return on Investment
RRP	Reciprocating Rod Pumping
R+D	Research & Development
TBG	Tubing
USA	United States of America
XL	Extra Large Rotor
bbls	Barrels
$Ca_2B_6O_{11}.5H_2O$	Colemanite
CO_2	Carbon Dioxide
C276	Hastelloy C276
ft	Feet
ft*lbs	Torque in foot pounds
H_3BO_3	Boric acid
H_2S	Hydrogen Sulfide
in	inch
m	Meter
m^3	Cubic meter
mm	millimeter
$Na_2B_4O_5(OH)_4.8H_2O$	Borax
$Na_2B_4O_6(OH)_2.3H_2O$	Kernite
$Na_2B_4O_7{\cdot}10H_2O$	Heating Borax
$NaCaB_5O_9.8H_2O$	Ulexite
≈	Almost Equal to
%	Percent or Percentage
°C	Celsius degree
°F	Fahrenheit degree

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